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$$\mathbf{f} : \Omega \subseteq \mathbb{R} \longrightarrow \mathbb{R}^4 : x \mapsto (2^{\sec x},\ \coth x,\ \log_5 \sqrt{\cos x},\ x^x)$$

$$D\mathbf{f}(x) = \begin{pmatrix} \dfrac{\partial f_1}{\partial x} \\ \dfrac{\partial f_2}{\partial x} \\ \dfrac{\partial f_3}{\partial x} \\ \dfrac{\partial f_4}{\partial x} \end{pmatrix}$$

- $\dfrac{\partial f_1}{\partial x} = 2^{\sec x} \ln 2 \cdot \sec x \tan x$
- $\dfrac{\partial f_2}{\partial x} = -\dfrac{1}{\sinh^2 x}$
- $f_3 = \log_5 \sqrt{\cos x} = \dfrac{\ln(\cos x)}{2 \ln 5}$

  $\dfrac{\partial f_3}{\partial x} = \dfrac{1}{2\ln 5} \cdot \dfrac{-\sin x}{\cos x} = -\dfrac{\tan x}{2 \ln 5}$
- $f_4 = x^x = e^{x \ln x}$

  $\dfrac{\partial f_4}{\partial x} = x^x(\ln x + 1)$

$$\Rightarrow D\mathbf{f}(x) = \begin{pmatrix} 2^{\sec x} \ln 2 \cdot \sec x \tan x \\ -\dfrac{1}{\sinh^2 x} \\ -\dfrac{\tan x}{2 \ln 5} \\ x^x(\ln x + 1) \end{pmatrix}$$

References